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# **Involvement and the SMILE Model: A Dual Lens Framework for Social Media Spoilers**

*Short Paper*

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## **Abstract**

*Social media spoilers are increasingly shaped by platform affordances, algorithmic curation, and real-time participation rather than by interpersonal etiquette alone. This study develops a dual-lens framework integrating involvement theory and the SMILE model to examine how users cognitively and affectively respond to spoiler exposure on social media. The study conceptualizes spoiler encounters as cognitively appraised disruption events involving spoiler sensitivity, perceived severity, and perceived control, alongside emotional responses of anger and anxiety. Media form and timeliness are further proposed as contextual moderators shaping these relationships. An explanatory sequential mixed-methods design is adopted, combining a controlled online experiment with semi-structured interviews. The study contributes to research on social media literacy, platform governance, and algorithmic mediation by explaining why identical spoiler content may generate divergent reactions across users and release contexts.*

**Keywords:** spoilers, social media literacy, personal involvement, platform governance.

## **Introduction**

Spoilers on social media are not simply breaches of etiquette. They are socio-technical outcomes produced through platform design, algorithmic curation, and rapid interpersonal diffusion (Boyd-Graber et al., 2013). Ranking systems, push notifications, trending modules, and repost dynamics can expose users to revelatory content before they can adjust settings or avoid discussion spaces. This helps explain why prior findings on spoiler effects remain mixed across narrative forms, spoiler types, and outcome measures (Daniel & Katz, 2019; Johnson & Rosenbaum, 2015; Leavitt & Christenfeld, 2011; Levine & Raines, 2025). It also helps explain why spoiler effects vary across release periods and patterns of audience access (Li et al., 2022; Ryoo et al., 2021).

Yet spoiler exposure does not affect all users equally. Users differ in how strongly they value narrative uncertainty, how they interpret spoiler encounters, and how much control they believe they retain over platform exposure. This study therefore centers personal involvement as a key determinant of spoiler responses. Here, personal involvement refers to users' cognitive and affective investment in a narrative universe. Involvement theory suggests that highly involved individuals process relevant information more elaborately and experience it more intensely because the content is more self-relevant (Celsi & Olson, 1988; Zaichkowsky, 1985). Prior spoiler research similarly indicates that highly involved users are more sensitive to spoilers and more likely to react strongly to them (Johnson & Rosenbaum, 2015; Ryoo et al., 2021).

However, involvement alone cannot fully explain why users with similar levels of investment may interpret the same spoiler differently. To address this limitation, the study integrates involvement theory with the SMILE model and related scholarship on critical social media literacy (Cho et al., 2024; Schreurs & Vandenbosch, 2021; Valle et al., 2025). This literacy-based perspective emphasizes how users cognitively appraise platform-mediated encounters and evaluate their ability to manage online information environments. In spoiler contexts, these appraisals include spoiler sensitivity, perceived severity, and perceived control, which together shape the affective responses that follow. This framework helps explain why spoiler exposure may be interpreted either as a manageable inconvenience or as a significant disruption to narrative experience, even among similarly involved audiences.

Media form and timeliness further shape how these processes unfold. Serial narratives tend to heighten spoiler sensitivity because they depend on cumulative uncertainty and deferred resolution, whereas more episodic or self-contained narratives often reduce the consequences of a single revelation (Leavitt & Christenfeld, 2011; Levine & Raines, 2025; Mittell, 2007). Timing also matters. Spoilers encountered soon after release are more likely to be judged inappropriate or harmful, whereas later disclosures are more likely to be treated as legitimate discussion (Li et al., 2022; Mäkelä et al., 2025; Ryoo et al., 2021). These dynamics are especially visible during high-profile release periods, when engagement-driven ranking systems intensify spoiler circulation and users adopt indirect wording or algospeak that weakens static keyword defenses (Gillespie, 2018; Steen et al., 2023).

Platforms increasingly treat spoilers as design and governance issues rather than as matters of individual courtesy alone. Current responses include keyword muting, temporary snoozing, and blur-to-reveal overlays (Mäkelä et al., 2025; Romaguera, 2022). Yet these measures remain fragile because they depend on user initiative, correct tagging, and filters that can recognize rapidly changing spoiler language. Studios and fan communities have also responded strategically during major release periods through controlled leaks, decoys, and coordinated messaging, further demonstrating that spoiler management has become embedded within platformed release culture rather than remaining a purely interpersonal matter (Boucher, 2019; Pell, 2024).

From this perspective, spoiler governance is best understood as an interaction among platform accountability, user expectations, and narrative timing. Platform governance scholarship emphasizes transparency, predictability, fairness, and legitimacy in online content regulation (Gillespie, 2018; Gorwa, 2019), while human-computer interaction research shows that spoiler exposure frequently occurs during live or peak-attention moments when etiquette alone cannot scale and automated intervention becomes necessary (Boyd-Graber et al., 2013; Mäkelä et al., 2025; Ryoo et al., 2021). Design choices such as delaying likely spoiler posts, adding time-bound overlays, or down-ranking high-risk content may preserve narrative integrity for some users while constraining immediacy for others. Their effects are therefore likely to depend on users' level of involvement, the form of the media, and the timing of exposure. Accordingly, this study asks: ***How do media form and timeliness moderate the relationship between personal involvement and users' cognitive and affective responses to spoilers on social media?*** To address this question, the study adopts an explanatory sequential mixed-methods design in which a quantitative online experiment is followed by qualitative interviews. This design allows the study to identify patterned effects and then interpret how users negotiate spoiler norms, platform affordances, and the tension between real-time participation and narrative protection.

## **Conceptual Background and Research Framework**

### ***Spoilers in Social Media: Platform Accountability and User Immediacy***

On social media, spoilers function as socio-technical processes rather than isolated acts of disclosure. Their consequences emerge from the interaction of interpersonal sharing, engagement-driven amplification, and narrative interpretation. During premieres, finales, and other high-attention release windows, these processes can increase the visibility of revelatory content, speed its circulation, and intensify perceptions of norm violation (Levine et al., 2016; Mäkelä et al., 2025; Völcker, 2017). Spoiler exposure is therefore shaped not only by what users post but also by how platforms rank, recommend, notify, and temporally organize content.

This produces a structural tension between platform accountability and user immediacy. Platform accountability refers to the rules, disclosures, interface cues, and ranking adjustments through which platforms govern revelatory content. User immediacy refers to the expectation of synchronous, low-friction participation that social media encourages through live discussion, notifications, and engagement incentives (Gillespie, 2018; Gorwa, 2019; Steinert & Dennis, 2022). Existing anti-spoiler tools such as mute filters, blur gates, and keyword restrictions restore some user agency, but they remain limited by tagging gaps, ranking overrides, and evasive language practices such as algospeak (Clune & McDaid, 2024; Mäkelä et al., 2025; Steen et al., 2023). As a result, spoiler governance cannot rely only on user-side customization. It also requires platform-side detection and intervention that are sensitive to context, timing, and evolving language.

Recent computational work points in this direction. Spoiler cues can be detected in text and, increasingly, in visual content, while interface approaches such as information clouding can reduce accidental exposure while preserving optional access (Boyd-Graber et al., 2013; Kinoshita et al., 2024; Nakamura & Komatsu, 2012; Tran et al., 2023). These approaches support time-sensitive interventions, including down-ranking predicted spoilers, applying temporary spoiler overlays, and offering opt-in reveals during high-risk release windows. Still, technical performance alone is not enough. Platforms must also decide when intervention is justified, how long it should last, and how much friction users will accept in exchange for spoiler protection. Spoilers therefore provide a useful case for examining platform governance under conditions where timing, relevance, and user expectations are all contested.

### ***Involvement and Social Media Literacy: An Integrated Lens on Spoilers***

Although spoilers are shaped by platform conditions, they are also interpreted through individual-level psychological processes. Personal involvement is central because it determines how much motivational relevance users attach to a narrative object. Highly involved users are more likely to process spoiler-related information concretely and emotionally, whereas low-involvement users are more likely to rely on more cursory or heuristic judgments (Celsi & Olson, 1988; Zaichkowsky, 1985). In spoiler contexts, this difference matters because the same disclosure can register as trivial background information for one user and as a serious disruption for another.

The present study treats personal involvement as the primary independent variable because it captures relatively stable motivational relevance. By contrast, media form and timeliness are treated as contextual moderators. This distinction is consistent with involvement research that separates enduring involvement from situational conditions that intensify or weaken involvement responses (Richins et al., 1992; Zaichkowsky, 1985). In the current setting, narrative form structures what counts as a meaningful revelation, and timeliness structures whether that revelation is judged acceptable, premature, or harmful. Serial narratives, for example, often heighten dependency across episodes and increase sensitivity to future-revealing information, while timing effects are strongest when access remains uneven in early release windows (Johnson & Rosenbaum, 2015; Li et al., 2022; Mittell, 2007).

However, involvement theory does not fully explain why users with similar levels of investment can respond differently to the same spoiler. To address this, the study incorporates the SMILE model, which conceptualizes social media literacy as a set of intertwined cognitive and affective structures that shape how users interpret, evaluate, and regulate algorithmically curated content (Schreurs & Vandenbosch, 2021). This extension is especially useful in spoiler settings because spoilers are not only pieces of content; they are interpreted as more or less revealing, more or less norm-violating, and more or less manageable.

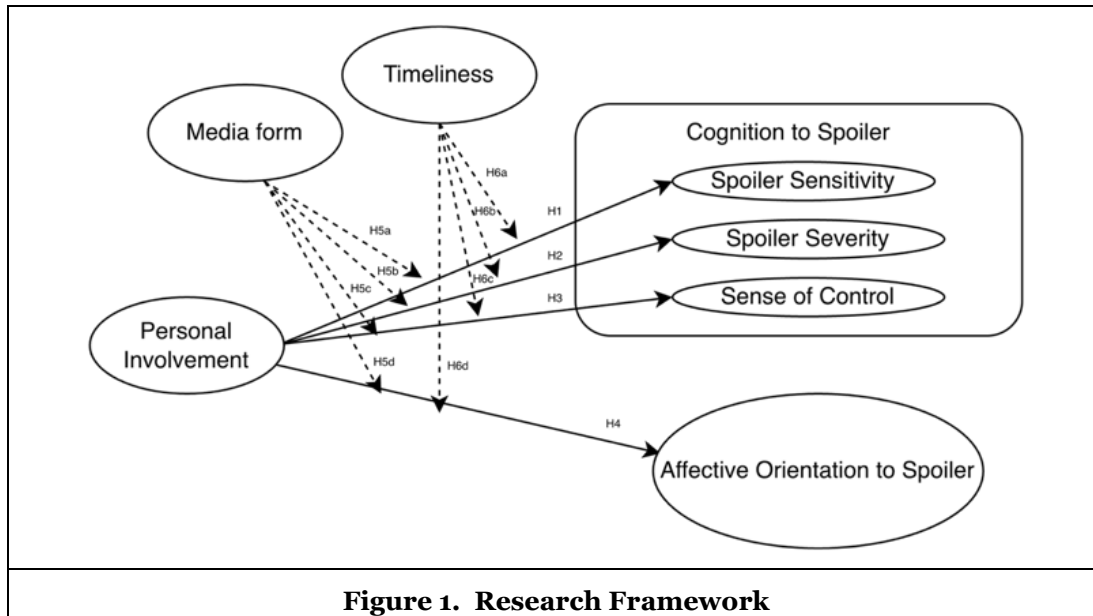
Drawing on the SMILE model, appraisal theory, and related social media literacy scholarship, this study conceptualizes spoiler-related literacy through both cognitive and affective responses (Cho et al., 2024; Lazarus, 1991; Schreurs & Vandenbosch, 2021; Smith & Ellsworth, 1985; Valle et al., 2025). When users encounter spoilers on social media, they are understood to make three immediate cognitive appraisals in sequence: first, how disruptive the disclosure is to the narrative experience; second, whether the disclosure is appropriate within a specific social and temporal context; and third, whether unwanted exposure can be managed or avoided. Thus, the cognitive dimension is operationalized through the three appraisals most directly activated during spoiler exposure: spoiler sensitivity, perceived severity, and perceived control over exposure. Spoiler sensitivity captures users' recognition of spoilery content and its appropriateness in relation to timing expectations. Perceived severity captures the degree to which exposure is judged harmful or disruptive to narrative experience. Perceived control captures users' perceived ability to regulate or avoid

spoiler exposure through platform tools and self-regulation strategies. The affective response refers to emotional intensity, operationalized here through anger and anxiety.

Together, involvement theory and the SMILE model recast spoilers as literacy-mediated psychological events embedded in algorithmic systems. This integrated lens explains why identical spoiler posts can generate divergent judgments across users and across time. It also allows spoiler research to contribute to broader information systems discussions of algorithmic mediation, governance, user agency, and digital literacy.

### Research Model and Hypotheses

Figure 1 presents the research model. Personal involvement is expected to influence how users identify, evaluate, and react to spoilers on social media. Within the integrated framework, spoiler encounters activate cognitive responses in the form of spoiler sensitivity, perceived severity, and perceived control, as well as affective responses in the form of emotional intensity.



**Figure 1. Research Framework**

Highly involved users possess richer narrative knowledge and attach greater self-relevance to the content they follow. Involvement theory suggests that personally relevant content receives greater attention and more elaborate processing (Celsi & Olson, 1988; Zaichkowsky, 1985). In spoiler contexts, this means that highly involved users are not only more knowledgeable about characters, plotlines, and release schedules, but also more motivated to monitor narrative-relevant cues. They are therefore more likely to notice potentially revelatory information and to classify ambiguous or indirect posts as spoilers. Personal involvement should therefore increase spoiler sensitivity.

**H1:** Personal involvement positively influences spoiler sensitivity.

Once content is recognized as a spoiler, highly involved users should also be more likely to judge it as harmful. Their stronger expectations, greater attachment to the narrative, and higher stakes in preserving uncertainty make disruption more consequential for them than for casual viewers (Johnson & Rosenbaum, 2015). Although some highly engaged fans may occasionally welcome controlled leaks or advance information as part of participatory culture (Völcker, 2017), the overall expectation is that stronger involvement heightens perceived spoiler severity.

**H2:** Personal involvement positively influences perceived spoiler severity.

Personal involvement may also increase perceived control over spoiler exposure. Highly involved users are more likely to seek out protective strategies, learn platform settings, and actively manage their media environment during release periods. Even if actual control remains imperfect, their greater effort and awareness may strengthen perceived control (Mäkelä et al., 2025).

**H3:** Personal involvement positively influences perceived sense of control.

Personal involvement may also increase perceived control over spoiler exposure. Although highly involved users may face greater exposure risk because they follow more related accounts and engage more frequently with relevant content, their stronger motivation to protect the viewing experience may also lead them to develop more active management strategies. They may be more likely to learn platform settings, mute keywords, avoid high-risk discussion spaces, monitor release windows, and temporarily adjust their media routines. From a perceived control perspective, these accumulated strategies can strengthen users' confidence in their ability to manage unwanted exposure, even when actual control remains imperfect (Bandura, 1997; Mäkelä et al., 2025). Personal involvement should therefore increase perceived sense of control.

**H4:** Personal involvement positively influences affective orientation to spoilers (emotional intensity).

The strength of these relationships is unlikely to be constant across narrative forms. Media form matters because it changes the narrative value of withheld information. Episodic narratives often distribute meaning locally, with conflicts and revelations resolved within discrete installments, which can reduce the consequences of a single disclosure. By contrast, serial or arc-based narratives rely more heavily on cumulative uncertainty, delayed resolution, speculation, and emotional payoff (Mittell, 2007). Under these conditions, personal involvement should translate more strongly into spoiler sensitivity, because highly involved users have more narrative cues to monitor; perceived severity, because revelations threaten longer-term narrative investment; perceived control, because users have stronger incentives to manage exposure; and affective orientation, because disrupted anticipation is more emotionally consequential. Therefore, the effects of personal involvement should be stronger for serial or arc-based narratives than for episodic narratives (Johnson & Rosenbaum, 2015; Mittell, 2007).

**H5:** Media form moderates the relationship between personal involvement and (a) perceived spoiler severity, (b) spoiler sensitivity, (c) perceived sense of control, and (d) affective orientation to spoilers.

Timeliness should likewise shape the strength of involvement effects. Spoilers encountered shortly after release are more likely to be interpreted as norm violations because access remains uneven and uncertainty remains central to the viewing experience. As a release window matures and access broadens, the same information is more likely to be treated as routine discussion. Accordingly, the effect of involvement on spoiler-related responses should be stronger during early release windows than during later ones (Li et al., 2022).

**H6:** Timeliness moderates the relationship between personal involvement and (a) perceived spoiler severity, (b) spoiler sensitivity, (c) perceived sense of control, and (d) affective orientation to spoilers.

## **Research Methodology**

This study adopts an explanatory sequential mixed-methods design in which a quantitative experiment is followed by qualitative interviews to deepen interpretation of the initial results (Creswell & Plano Clark, 2018; Fetter et al., 2013; Ivankova et al., 2006). The quantitative phase identifies how personal involvement, media form, and timeliness shape spoiler responses. The qualitative phase then explains how participants interpret those patterns in practice, particularly how they negotiate spoiler norms, platform affordances, and the tension between real-time participation and narrative protection. This design is appropriate because the study seeks both explanation and interpretation: it first tests patterned relationships and then examines how users make sense of those relationships in social media settings.

Integration occurs at multiple points in the design. The quantitative findings inform interview sampling and the construction of the interview guide, while the qualitative findings are used to explain unexpected patterns, clarify boundary conditions, and refine interpretation of the moderating roles of media form and timeliness. In this way, the qualitative phase is not a separate add-on. It is explicitly connected to the experiment and functions as the interpretive extension of the quantitative phase.

## **Proposed Data Collection**

The quantitative phase uses a controlled online experiment to examine how personal involvement, media form, and timeliness shape cognitive and affective responses to spoilers on social media. Participants will

be recruited through social media platforms and university networks. Eligibility will be limited to individuals who regularly consume films or series and who have encountered at least one spoiler on social media during the past year. Respondents with minimal engagement in media consumption or social media use will be excluded because they are unlikely to provide meaningful evaluations of spoiler scenarios.

The experiment employs a mixed factorial design, with media form (film vs. series) manipulated between subjects and timeliness (early vs. late spoiler) manipulated within subjects. Each participant is randomly assigned to either the film condition or the series condition and is exposed only to spoiler content corresponding to that media type. Within that assigned media type, participants encounter both timeliness conditions: spoiler content presented as occurring shortly after release and spoiler content framed as occurring several weeks after release. This design allows the study to compare temporal sensitivity within the same individual while testing whether spoiler responses differ across media forms.

At the platform entry point, a pre-test module assesses participants' familiarity with and anticipated engagement in the target IP to ensure that the spoiler constitutes a meaningful informational threat. Specifically, the study will select an upcoming film or series title scheduled for release that has already received substantial public exposure, such as official trailers, platform promotions, or high social-media visibility. The pre-test will verify participants' baseline awareness or familiarity with the title and their anticipated viewing interest so that spoiler exposure constitutes a meaningful informational threat rather than a neutral disclosure. Based on the selected title, the experiment will employ researcher-crafted but plausible plot spoilers. During the pre-test, spoiler stimuli will be evaluated for perceived plausibility and perceived "spoiler-ness," meaning whether they are viewed as revealing important plot information, and only stimuli meeting predefined thresholds will be retained for the main study.

Participants interact with a researcher-developed mock platform resembling contemporary feed-based social media environments, such as Threads or X. The interface includes user avatars, usernames, relative timestamps, post captions, and interactive affordances such as like, comment, and share functions. This design enhances ecological validity by approximating natural browsing behavior and reducing artificiality in stimulus presentation. Within each phase, participants browse a simulated feed containing target spoiler posts embedded among unrelated posts, such as food, daily life updates, or weather-related content, ensuring that spoiler exposure occurs incidentally during browsing.

After providing informed consent and demographic information, participants complete the personal involvement measure before entering the experimental task. Personal involvement is measured using an adapted version of Zaichkowsky's (1985) Personal Involvement Inventory. Following each spoiler exposure condition, participants report immediate cognitive and affective responses. Cognitive responses include spoiler sensitivity, perceived severity, and perceived control over spoiler exposure, while affective responses focus on emotional intensity operationalized through anger and anxiety. These measures draw on prior social media literacy and emotion research, including Schreurs and Vandenbosch (2021), PANAS anger items (Vera-Villarreal et al., 2019), and short-form state anxiety items from the STAI-S (Zsido et al., 2020). Although spoilers may occasionally generate curiosity or anticipation (Leavitt & Christenfeld, 2011), spoilers on social media are more commonly experienced as disruptions of narrative expectations and timing norms. Demographic and media-use variables, including age, gender, occupation, platform use, and viewing habits, will therefore be collected as controls to isolate the effects of personal involvement, media form, and timeliness.

After the experiment, a subset of participants will be invited to participate in semi-structured interviews designed to explain and contextualize the quantitative findings. Consistent with explanatory sequential mixed-methods logic, interview sampling will follow a purposive extreme-case strategy guided by notable quantitative response patterns (Creswell & Plano Clark, 2018; Fetter et al., 2013). Specifically, participants will be selected based on predefined criteria, including responses that deviate substantially from sample means (e.g., unusually high or low spoiler sensitivity, perceived severity, perceived control, or emotional intensity), unexpected combinations of responses across variables, or patterns associated with significant interaction effects between personal involvement, media form, and timeliness. These criteria are intended to identify participants whose responses may help explain underlying cognitive and emotional mechanisms that are not fully interpretable from quantitative scores alone. Interview questioning will then probe how participants interpret spoiler legitimacy, regulate exposure, and negotiate platform norms under different spoiler conditions.

## **Proposed Data Analysis**

Quantitative data will be analyzed in two stages. First, the measurement properties of the multi-item scales will be examined through reliability and dimensionality checks before hypothesis testing, which is appropriate when established instruments are adapted to a new empirical context (Brown, 2015; Kline, 2023). Second, the hypotheses will be tested using linear mixed-effects models because each participant evaluates four scenarios and the resulting observations are therefore nested within individuals. Mixed-effects models are well suited to repeated-measures data because they account for within-person dependence, accommodate both participant-level and condition-level predictors, and offer greater flexibility than traditional repeated-measures ANOVA (Baayen et al., 2008; Gueorguieva & Krystal, 2004; Hesser, 2015).

After measurement quality is confirmed, scale scores based on the validated measurement model will be computed for spoiler sensitivity, perceived severity, perceived control, anger, and anxiety, and separate mixed-effects models will be estimated for each dependent variable (Brown, 2015; Kline, 2023). Personal involvement will be modeled as a participant-level predictor, whereas media form and timeliness will be modeled as within-participant experimental factors. The main tests concern the interaction terms between personal involvement and each manipulated factor, as these directly correspond to the moderation hypotheses. Random intercepts for participants will be specified, and random slopes for the within-subject factors will be retained where supported by the design and model convergence, because the random-effects structure should reflect the experimental design as closely as the data allow (Baayen et al., 2008; Barr et al., 2013). Significant interactions will then be probed to clarify how the effect of personal involvement varies across media-form and timing conditions.

Qualitative interview data will be analyzed using reflexive thematic analysis, and the findings will be interpreted alongside the quantitative results to identify convergence, divergence, and explanatory nuance across the two phases. This form of integration is consistent with explanatory sequential mixed-methods designs, in which qualitative findings are used to explain or elaborate quantitative patterns (Braun & Clarke, 2006, 2021; Creswell & Plano Clark, 2018; Fetters et al., 2013).

## **Expected Contributions**

This research-in-progress study conceptualizes spoilers as literacy-mediated psychological events shaped by user involvement and social media literacy rather than by content alone. By integrating involvement theory with the SMILE model, it clarifies how motivational relevance shapes spoiler sensitivity, severity judgments, perceived control, and emotional intensity. The inclusion of media form and timeliness further explains why spoiler responses vary across narrative structures and release windows. In this way, the study places spoiler perception within broader information systems discussions of algorithmic mediation, platform governance, and user agency.

The study also has practical implications for platform design. If spoiler responses are strongest among highly involved users and during early release windows, platforms could implement more adaptive anti-spoiler protections during these periods. These protections might include time-bound overlays, media-sensitive spoiler gates, and filtering systems that respond to both semantic signals and release-stage context. Such mechanisms would not eliminate discussion, but they could better balance participation with protection. The study therefore offers a basis for designing anti-spoiler interventions that are more responsive to user involvement, more sensitive to narrative timing, and more effective in reducing unintended spoiler exposure.

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